

Diabetes Mellitus

What is Diabetes?

Diabetes is a chronic disease where the body cannot properly regulate blood sugar (glucose). Type 1 diabetes occurs when the body produces no insulin. Type 2 diabetes is when the body does not use insulin effectively.

Gestational diabetes occurs during pregnancy. Pre-diabetes is when blood sugar is high but not yet in the diabetic range.

Symptoms

Type 1 symptoms appear suddenly: frequent urination, excessive thirst, extreme hunger, unexplained weight loss, fatigue, blurry vision, slow-healing sores.

Type 2 symptoms develop slowly and may include the same signs plus frequent infections, numbness or tingling in hands and feet (neuropathy).

Diabetic ketoacidosis (DKA) signs: fruity breath, nausea, vomiting, abdominal pain, confusion — this is an emergency.

Medicines

Insulin (rapid, short, intermediate, long-acting) is required for Type 1 and some Type 2 patients.

Metformin is the first-line oral medication for Type 2 diabetes. It reduces glucose production in the liver.

Sulfonylureas (glipizide, glibenclamide) stimulate the pancreas to produce more insulin.

SGLT2 inhibitors (empagliflozin, dapagliflozin) cause kidneys to remove extra glucose via urine.

GLP-1 receptor agonists (semaglutide, liraglutide) help control blood sugar and promote weight loss.

Emergency Actions

Hypoglycemia (low blood sugar below 70 mg/dL): give 15g fast-acting carbs (glucose tablets, juice, sugar). Recheck in 15 minutes. Call emergency if unconscious.

Hyperglycemia emergency (DKA): seek emergency care immediately. Do not attempt home treatment.

If unconscious and hypoglycemia suspected: do not give food/drink by mouth. Call 112/911. Give glucagon injection if available.

Keep glucose tablets or juice accessible at all times.

What You Should Do

Monitor blood glucose regularly as advised by your doctor (fasting and post-meal readings).

Follow a balanced diet with controlled carbohydrates, high fiber, and low sugar.

Exercise 150 minutes per week of moderate activity to improve insulin sensitivity.

Take all medications on schedule. Never skip insulin doses.

Get regular eye, kidney, and foot checkups. Inspect feet daily for cuts or wounds.